LITERATURE & METHODOLOGY

PROBLEM IDENTIFICATION

Rapid urban growth and fragmented multimodal integration have led to temporal inequities, unreliable transfers, and hidden travel burdens in Chennai's public transport system. Limited synchronization between metro, suburban rail, and bus networks results in long transfer waits, missed connections, and unequal accessibility, while existing frameworks overlook reliability and transfer penalties. This calls for a Transfer Friction–based Integrated Transit Reliability Framework that quantifies inefficiencies and addresses spatial inequities through GTFS-based temporal modeling.

FOCUS AREAS

- Multimodal Network Integration
- Transfer Friction & Temporal Reliability
- Accessibility & Equity Assessment
- GTFS-based Time-Expanded Network Modeling
- Urban Transport Governance
- Spatial Inequality & Mobility Justice

Literature Review

CASE AREA: CHENNAI

Research Question

- How does transfer friction vary across Chennai's multimodal network?
- Which socio-spatial groups experience the highest temporal inequity?
- How reliable are multimodal transfers under schedule-based planning?
- Can GTFS-based probabilistic modeling improve accessibility planning?
- How can Transfer Friction Index guide policy decisions?

LITERATURE REVIEW

Title	Author(s)	Journal / Year	Objective	Relevance
Travel Time Reliability & Valuation	Bates et al.	TR Part E, 2001	Reliability distribution framework	Foundation for reliability buffer time
Accessibility Theory	Geurs & van Wee	TR Part B, 2004	Accessibility components	Theoretical base for accessibility metrics
GTFS Network Modeling	Fortin et al.	JTG, 2016	Graph-based GTFS analysis	Methodological backbone
Transfer Penalty Estimation	Navarrete & Ortúzar	TR Part A	Transfer disutility modeling	Supports friction quantification
Train Delay Reliability	Stratil-Sauer & Breyer	JPT, 2026	Probabilistic transfer modeling	Algorithmic structure for reliability modeling
Transfer Friction & Spatiotemporal Networks	Liu et al.	2025	GTFS time-expanded modeling	Direct support for network modeling
Convenience Valuation	OECD ITF	2014	User perception metrics	Behavioral dimension
Time Geography & Individual Accessibility	Neutens et al.	2007	Temporal constraints	Supports temporal inequity concept

KEY TAKEAWAYS

- Accessibility must incorporate reliability and transfer friction, not just travel time.
- Temporal inequity is spatially clustered.
- GTFS-based probabilistic modeling enables dynamic accessibility evaluation.
- Transfer Friction Index provides measurable policy leverage.
- Multimodal integration requires synchronization, not just infrastructure expansion.

AIM

To develop a Transfer Friction Index (TFI)-based Integrated Multimodal Reliability Framework to quantify temporal inequity and improve accessibility in Chennai's public transport system.

HYPOTHESIS: Implementing a Transfer Friction–informed multimodal integration framework can significantly reduce temporal inequity and improve accessibility reliability across socio-economic groups in Chennai.

OBJECTIVES

To analyze multimodal network structure and transfer dependencies.

- **Map GTFS network (Metro, Suburban, Bus)**
- **Identify major interchange nodes**
- Model time-expanded network graph
- Analyze urban demographics of wards

- GTFS Data (All Modes: Metro, MTC Bus, Suburban Rail) agency.txt, routes.txt, trips.txt, stop_times.txt, stops.txt, calendar.txt, calendar_dates.txt, transfers.txt, shapes.txt, frequencies.txt
- Spatial Data - GeoJSON

Spatial analysis (Primary Data & Secondary Data)

INTENDED ANALYSIS: 1. Graph connectivity analysis, 2. Betweenness centrality, 3. Transfer density mapping

To quantify Transfer Friction and reliability patterns.

- **Measure transfer waiting times**
- **Estimate missed connection probability**
- Compute reliability buffer time (95–50 percentile)

- Temporal Schedule Data :- Arrival times, Departure times, Headways (derived from stop_times), Peak vs off-peak service patterns
- Operational Data :- Historical delay records, Real-time arrival deviation data

Spatial analysis (Primary Data & Secondary Data)

INTENDED ANALYSIS: 1. Probabilistic delay modeling, 2. Transfer penalty calibration, 3. Reliability rating computation

To identify friction hotspots and critical interchange nodes.

- **High missed-transfer clusters**
- **Long cumulative wait zones**
- High vulnerability–low accessibility overlap

- Spatial Layers :- Interchange station locations, Ward boundaries (GeoJSON/KML), Zone boundaries CMA boundary
- Urban Context Data :- Land-use data, Employment hubs, Commercial centers

Spatial analysis (Primary Data & Secondary Data)

INTENDED ANALYSIS: 1. Hotspot (Getis-Ord Gi*), 2. Network resilience assessment, 3. Sensitivity analysis

To assess spatial-temporal inequity in accessibility.

- **Ward-level accessibility variation**
- **Travel time variability mapping**
- Socioeconomic vulnerability overlay

- Demographic Data (Ward Level) :- Total population, Population density, Worker population, Income proxy indicators, Slum population, Vehicle ownership, Gender ratio

Spatial analysis (Primary Data & Secondary Data)

INTENDED ANALYSIS: 1. Isochrone comparison (Peak vs Off-Peak), 2. Gini coefficient of accessibility, 3. Spatial regression analysis

To propose a policy-integrated Transfer Friction Reduction Framework.

- **Timetable synchronization strategies**
- **Interchange design improvements**
- Dynamic transfer buffer policies & informed journey planning

- Policy Data :- Integration policy documents (CMRL, MTC, SR), Fare structure data, Timetable coordination, policies TOD guidelines

Secondary Data) For understand current policy measures

INTENDED ANALYSIS: 1. Scenario modeling, 2. Cost-benefit comparison, 3. Impact simulation